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Trichome structure and evolution in Neotropical lianas

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• *Background and Aims* Trichomes are epidermal outgrowths generally associated with protection against herbivores and/or desiccation that are widely distributed from ferns to angiosperms. Patterns of topological variation and morphological evolution of trichomes are still scarce in the literature, preventing valid comparisons across taxa. This study integrates detailed morphoanatomical data and the evolutionary history of the tribe Bignonieae (Bignoniaceae) in order to gain a better understanding of current diversity and evolution of trichome types.
• *Methods* Two sampling schemes were used to characterize trichome types: (1) macromorphological characterization of all 105 species currently included in Bignonieae; and (2) micromorphological characterization of 16 selected species. Individual trichome morphotypes were coded as binary in each vegetative plant part, and trichome density and size were coded as multistate. Ancestral character state reconstructions were conducted using maximum likelihood (ML) assumptions.
• *Key Results* Two main functional trichome categories were found: non-glandular and glandular. In glandular trichomes, three morphotypes were recognized: peltate (*Pg*), stipitate (*Sg*) and patelliform/cupular (*P/Cg*) trichomes. Non-glandular trichomes were uniseriate, uni- or multicellular and simple or branched. *Pg* and *P/Cg* trichomes were multicellular and non-vascularized with three clearly distinct cell layers. *Sg* trichomes were multicellular, uniseriate and long-stalked. ML ancestral character state reconstructions suggested that the most recent common ancestor (MRCA) of Bignonieae probably had non-glandular, *Pg* and *P/Cg* trichomes, with each trichome type presenting alternative histories of appearance on the different plant parts. For example, the MRCA of Bignonieae probably had non-glandular trichomes on the stems, prophylls, petiole, petiolule and leaflet veins while *P/Cg* trichomes were restricted to leaflet blades. *Sg* trichomes were not present in the MRCA of Bignonieae independently of the position of these trichomes. These trichomes had at least eight independent origins in tribe.
• *Conclusions* The patterns of trichome evolution indicate that most morphotypes are probably homologous in Bignonieae and could be treated under the same name based on its morphological similarity and common evolutionary history, in spite of the plethora of names that have been previously designated in the literature. The trichome descriptions presented here will facilitate comparisons across taxa, allowing inferences on the relationships between trichome variants and future studies about their functional properties.

Key words: Bignoniaceae, Brazil, EFNs, extrafloral nectaries, glands, insect–plant interactions, morphological evolution, trichomes, vines.

INTRODUCTION

A good understanding of the evolutionary patterns of individual morphological features represents the basis of comparative biology (MacLeod and Forey, 2002). Such studies combine information derived from phylogenetic analyses with detailed morphological data to improve understanding of the patterns of morphological change (e.g. Fougère-Danezan *et al.*, 2010). The identification of a common origin for individual morphological features represents a critical step for the establishment of homology hypotheses (Scotland and Pennington, 2000; West-Eberhard, 2003), which is essential for comparisons accross taxa (e.g. Fahn, 1986), and for the test of adaptive hypotheses (Larson and Losos, 1996).

Trichomes are epidermal outgrowths that are widely distributed in land plant groups (Levin, 1973; Payne, 1978). These structures are often associated with protection against herbivores and/or desiccation (Wagner *et al.*, 2004), and are of extreme importance for the maintenance of various plant groups in herbivore-rich environments and dry areas. Current patterns of trichome variation are thought to have resulted from natural selection, with individuals that have trichomes showing greater advantages (e.g. leaf damage, loss of water or high temperature) when compared with individuals lacking those structures (e.g. Levin, 1973; Elle *et al.*, 1999; Valverde *et al.*, 2001; Romero *et al.*, 2008; Johnson *et al.*, 2009; Kaplan *et al.*, 2009).

Directional selection (Valverde *et al.*, 2001) and stabilizing selection (Elle *et al.*, 1999) have often been used to explain the patterns of trichome variation encountered in natural populations. Such models have gained further support from genetic studies that have characterized DNA regions responsible for the patterns of trichome variation of model organisms [e.g.

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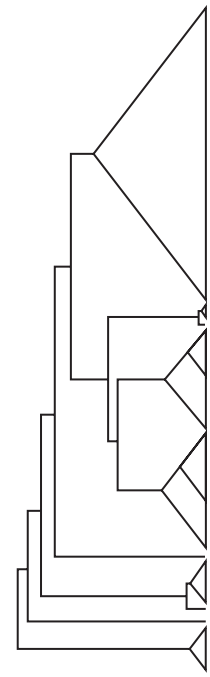
quantitative trait locus (QTL) studies by Mauricio, 2005; Symonds *et al.*, 2005]. Such studies have allowed the use of trichomes in population-level quantitative evolutionary studies. Unfortunately, little is still known about the general morphology of trichomes and about their patterns of topological variation (Theobald *et al.*, 1979), preventing valid comparisons across taxa. In addition, macroevolutionary patterns of trichome types based on robust phylogenetic analyses are still scarce (but see Beilstein *et al.*, 2006; Chauveau *et al.*, 2011) and little is still known regarding the mode, timing and pattern of evolution of trichome density in plants as a whole.

Trichomes in Bignoniaceae and tribe Bignonieae

Bignoniaceae exhibit notable variation in terms of trichome structure, position, abundance and size among species (e.g. Seibert 1948). Indeed, various studies have reported different trichome types in representatives of Bignoniaceae (see Fig. 1), and species exhibit a variety of glandular and non-glandular trichomes on different vegetative plant parts (e.g. Seibert, 1948) and reproductive organs (see Souza *et al.*, 2010). However, detailed structural studies are still lacking for most of these trichomes. Though glandular trichomes have been reasonably well documented in some species, sampling is still incomplete in the family (Fig. 1).

The lack of structural studies on different trichome types has prevented the establishment of hypotheses of common origin and/or meaningful comparisons among trichome morphotypes in Bignoniaceae. Furthermore, a wide array of terms have been used to describe the various types of glandular trichomes in Bignonieae (Fig. 1), complicating the distinction among trichome morphotypes and preventing evolutionary studies with trichomes in this group. For example, the patelliform glandular trichomes distributed across the leaf surface, petioles, prophylls of axillary buds, interpetiolar regions, calyces, corollas and fruits of representatives of the tribe have been called 'nectaries' or 'extrafloral nectaries' (e.g. Elias and Gelband, 1976; Oliveira and Leitão-Filho, 1987; Nogueira *et al.*, 2012*a*), 'scale-like trichomes as morphotype of EFNs' (e.g. Díaz-Castelazo *et al.*, 2005; Machado *et al.*, 2008) and 'glands' (e.g. Seibert, 1948; Gentry, 1974, 1980; Laroche, 1974; Lohmann, 2006), making a standardization of the trichome terminology extremely necessary in this group.

Apart from showing great variation in trichome structure and morphology, Bignonieae represents the most abundant and diverse clade of Bignoniaceae and also of Neotropical lianas (Lohmann, 2006). The tribe includes 21 monophyletic genera and approx. 400 species (nearly half of all species of Bignoniaceae) (Lohmann and Taylor, 2013), 25 % of which have been sampled in a molecular phylogenetic analysis based on plastid (*ndhF*) and nuclear (*PepC*) markers (Lohmann, 2006). The wide variation found in trichome morphology, topology and other quantitative traits in representatives of Bignonieae and the robust phylogenetic tree available make this tribe an excellent group for addressing trichome evolution. A uniform trichome terminology in Bignonieae would greatly improve comparative taxonomic, ecological and evolutionary research in this morphologically diverse clade of lianas.



Phylogenetic relationships in Bignoniaceae	Tribe	Species name	Types of trichomes described in each published paper	References
	BIGNONIEAE	*Lundia corymbifera; Callichlamys latifolia*	Glands	Seibert (1948)
		Anemopaegma chamberlaynii	Glands and glandular (lepidote) scale/Simple uniseriate trichomes and foliar nectaries	Seibert (1948); and Inamdar (1969)
		Anemopaegma arvense	Two kinds of glandular trichomes	Hyakutake & Grotta (1965)
		Adenocalymma comosum	Simple and glandular capitate hairs; and patelliform glands	Laroche (1974)
		Bignonia capreolata	Scalelike nectaries	Elias (1983)
		Adenocalymma marginatum; Amphilophium paniculatum; Anemopaegma flavum; Arrabidaea chica; Arrabidaea corallina; Arrabidaea selloi; Clytostoma binatum; Dolichandra cynanchoides; Macfadyena dentate; Macfadyena unguis-cati; Mansoa difficilis; Melloa quadrivalvis; Parabignonia chodatii; Pithecoctenium crucigerum; Pithecoctenium cynanchoides	Extranupcial nectaries	Rivera (2000)
		Amphilophium paniculatum; and *Mansoa hymenaea*	Scale-like trichomes (morphotype of EFNs)	Díaz-Castelazo *et al.* (2005)
		Pyrostegia venusta	Tector and two kinds of glandular trichomes	Duarte and Jurgensen (2007)
		Arrabidea brachypoda	Scale-like trichomes and peltate trichomes (extrafloral nectaries)	Machado *et al.* (2008)
		Memora nodosa	Glandular and non-glandular trichomes	Tresvenzon *et al.* (2011)
	2	*Bignonia illicium*	Extrafloral nectaries	Subramanian & Inamdar (1986*a*)
		Tabebuia serratifolia	Extrafloral nectaries	Thomas & Dave (1992)
		Tabebuia palluda; Parmentiera cerufera; Parmentiera venusta; Crescentia cujete.		Ogundipe & Wujek (2004)
		Tabebuia rosea	Scale-like trichomes	Díaz-Castelazo *et al.* (2005)
	3	*Spathodea campanulata*	Extrafloral nectaries	Rao (1926)
		Kigelia pinnata	Extrafloral nectaries	Subramanian & Inamdar (1986*b*)
		Markhamiaa obtussiflora; Markhamia lutea; Markhamia tomentosa; Newbouldia leavis; Stereospermum kunthianum; Stereospermum acuminatissimum; Kigelia africana; Haplophragma adenophylla; Rhodocolea racemosa; Spathodea campanulata.	Simple trichomes (small acicular, subulate, filiform and fasciculate); dendritic trichomes (simple and complex); and two kinds of glandular trichomes (peltate scales and capitate glands)	Ogundipe & Wujek (2004)
	4	*Tecoma capensis*	Glandular hairs and patelliform glands (extrafloral nectaries)	Parija & Samal (1936); Subramanian & Inamdar (1989); and Ogungipe & Wujek (2004)
	5	*Campsis radicans; Campsis grandiflora*	Extrafloral nectaries	Elias & Gelband (1975)
		Jacaranda mimosifolia	Capitate glandular hair	Seibert (1948) and Ogundipe & Wujek (2004)
		Jacaranda decurrens	Peltate scale-like glandular and tector trichomes	Zatta *et al.* (2009)

FIG. 1. Previous morphological studies of glandular and non-glandular trichomes in Bignoniaceae and the phylogenetic relationships described by Olmstead *el al.* (2009), in which the clades 2, 3, 4 and 5 are *Tabebuia* alliance, Palaeotropical clade, Tecomeae and Jacarandeae, respectively. The shaded area represents the anatomical studies in the tribe Bignonieae. All taxonomic names are shown in the Supplementary Data Table S1.

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We reviewed the literature on the morphology and anatomy of trichomes in representatives across Bignoniaceae to understand better the diversity of trichomes of the focal group, tribe Bignonieae. Additionally, we characterized trichome types in representatives of all 21 genera currently recognized in Bignonieae to fill the data gaps currently encountered in the literature on the family. For those taxa, detailed anatomical studies, macromorphological descriptions and quantitative analyses of trichome size and density were carried out. This information was then used as a basis for understanding macroevolutionary patterns of non-glandular and glandular trichomes in Bignonieae using a robust molecular phylogenetic tree for the group (Lohmann, 2006). The following questions were addressed. (1) How are the trichome morphotypes distributed in vegetative parts of representatives of Bignonieae, what is their overall structure and how are they characterized morphologically? (2) Are the morphological variants of each trichome morphotype homologous, or has each trichome morphotype evolved independently during the phylogenetic history of the group? (3) What evolutionary patterns are evident for each trichome morphotype on different plant parts? (4) What is the evolutionary pattern of trichome size and density in Bignonieae?

MATERIALS AND METHODS

Trichome characterization

To obtain a clear picture of the patterns of morphological variation encountered in trichome types in representatives of Bignonieae, we first reviewed all currently available trichome studies of Bignoniaceae (Fig. 1). We then used two complementary sampling schemes to characterize the individual trichome types present in the vegetative portions of representatives of Bignonieae. For that, we used a stereomicroscope and a broad-scale sampling scheme to characterize the macromorphology of trichomes in the 105 species of Bignonieae included in the phylogenetic anlysis of the tribe. This first characterization resulted in the definition of the main trichome morphotypes in the tribe. We then conducted detailed anatomical studies using scanning emission microscopy (SEM) and light microscopy on a smaller number of taxa (16 species of Bignonieae), selected to characterize the micromorphology of all major trichome morphotypes. These two sampling schemes led to a detailed description of the overall morphology of the individual trichome types in Bignonieae.

Macromorphological characterization. We used herbarium specimens to characterize macromorphological patterns of variation of trichome types in representatives of Bignonieae. For that, three specimens of each of the 104 species of Bignonieae included in the combined molecular phylogenetic analysis of Lohmann (2006) plus *Callichlamys latifolia* and *Perianthomega vellozoi* were sampled. This sampling scheme included representatives of all 21 genera of Bignonieae and a sample of taxa across different locations and habitats. Herbarium specimens were analysed using a stereomicroscope (Olympus SZ60; zoom range of 6·3:1). The following variables were recorded for each trichome type: (*a*) presence/absence; (*b*) size; and (*c*) density on stems, prophylls of axillary buds, petioles, petiolules, and adaxial and abaxial leaf surfaces (basal, middle and upper portions). Trichome density was sampled in three squares of 1 mm^2 each, located in the basal, middle and upper portions of leaflets. Larger trichomes such as the patelliform/cupular glandular trichomes were sampled in larger squares of 1 cm^2. The number of patelliform/cupular glandular trichomes (abundance data) was also recorded for the interpetiolar region of stems and the prophylls of the axillary buds, petioles and petiolules.

Micromorphological characterization. We used fixed materials in a micromorphological characterization of trichome types in 16 species of Bignonieae. The following taxa were sampled: (1) *Adenocalymma pedunculatum* (Vell.) L.G.Lohmann; (2) *Amphilophium crucigerum* (L.) L.G.Lohmann; (3) *Amphilophium parkeri* (DC.) L.G.Lohmann; (4) *Anemopaegma album* Mart. ex DC.; (5) *Anemopaegma scabriusculum* Mart. ex DC.; (6) *Bignonia prieurei* DC.; (7) *Cuspidaria sceptrum* (Cham.) L.G.Lohmann; (8) *Dolichandra unguis-cati* (L.) L.G.Lohmann; (9) *Fridericia triplinervia* (Mart. ex DC.) L.G.Lohmann; (10) *Lundia nitidula* DC.; (11) *Mansoa difficilis* (Cham.) Bureau & K.Schum.; (12) *Perianthomega vellozoi* Bureau; (13) *Pleonotoma albiflora* (Salzm. ex DC.) A.H.Gentry; (14) *Pyrostegia venusta* (Ker Gawl.) Miers; (15) *Stizophyllum riparium* (Kunth) Sandwith; and (16) *Tanaecium pyramidatum* (Rich.) L.G.Lohmann. Field collections were made in areas of savanna (Parque Estadual de Grão Mogol, Minas Gerais State/Brazil and Parque Nacional da Chapada Diamantina, Bahia State/Brazil) and rain forest (Reserva Florestal Adolpho Ducke, Amazonas State/Brazil).

Mature and young vegetative plant parts were fixed in 50 % FAA (Johansen, 1940) and stored in 70 % ethanol. For light microscopy, the material was dehydrated in an ethanol series and then embedded in 2-hydroxyethylmethacrylate (Leica Microsystems Inc., Heidelberg, Germany). Longitudinal sections of 6 μm were cut with a rotary microtome and stained with 0·05 % toluidine blue (pH 4·3) (O'Brien *et al.*, 1964). For SEM studies, the fixed material was dehydrated in an ethanol series and critical point dried. Samples were mounted on stubs and sputter-coated with gold. Observations were made with a Quanta 200 scanning electron microscope (Fei Company, Hillsboro, OR, USA).

Overall anatomical descriptions and trichome classification were based on Theobalde *et al.* (1979). Specific terminology used for the characterization of the patelliform/cupular glandular trichomes followed Elias (1983). Herbarium vouchers used in this study are listed in Supplementary Data, Specimens.

Phylogenetics and trichome evolution

Phylogenetic tree. A moderately to well-supported combined molecular phylogenetic analysis of Bignonieae [i.e. with 92 % of the nodes receiving ≥50 % maximum likelihood (ML) bootstrap support] based on plastid (*ndhF*) and nuclear (*PepC*) sequences is currently available for the group (Lohmann, 2006). The tree used here was the same as the combined molecular phylogenetic analysis of Lohmann (2006) except that *Pyrostegia dichotoma* was treated as a synonym of *Pyrostegia venusta* and excluded from the tree, and *Callichlamys latifolia* and *Perianthomega vellozoi* were included in order to represent the only two genera of Bignonieae that were absent in the combined analysis of Lohmann (2006). The phylogenetic position of *Callichlamys* and *Perianthomega* followed relationships

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recovered in the *ndhF* phylogenetic analysis of Lohmann (2006), which included a broader sampling of taxa than the combined molecular (*ndhF* plus *PepC*) analysis of Lohmann (2006). This phylogenetic tree was combined with fossil data to estimate the time of divergence of all nodes in Bignonieae (Lohmann *et al*., 2013), and the tree used here is the same as that used in the earlier study of evolution of extrafloral nectaries of Bignonieae (Nogueira *et al*., 2012*b*).

Character coding. Each of the four trichome morphotypes recorded for Bignonieae were coded as present/absent (binary coding) according to their occurrence on each plant part considered (stems, prophylls of axillary buds, petioles, petiolules, and adaxial and abaxial sides of leaflets). In addition, continuous measurements of trichome densities were transformed into discrete character states and coded as multistate characters. The total range of trichome density among the 105 species of Bignonieae was divided into three classes equally. Those three classes of the range were used as the three states of trichome density: (1) low; (2) intermediate; and (3) high. The same approach was applied to trichome sizes.

Ancestral character state reconstructions. We initially used information on trichome position, structure and ontogeny (gathered from the literature, Fig. 1) to generate hypotheses of homology for the various trichome types. We subsequently combined all the morphological information with data on the phylogenetic history of Bignonieae through ML ancestral character state reconstructions in order to verify whether the morphological similarity was based on common ancestry. The ML ancestral character state reconstructions of trichome morphotypes were implemented in Mesquite 2·74 (Maddison and Maddison, 2007). In all analyses, trichome characters were equally weighted and considered unordered. For binary traits, two evolutionary models were considered as applied by Ree and Donoghue (1999): an Mk1 model with one parameter, in which any particular change is considered equally likely, and an Mk2 model with two parameters that represent forward and backward rates of changes between character states. Models were chosen using the likelihood ratio test (LRT). The evolutionary history of multistate character states (resulting from the discretization of quantitative characters) was only reconstructed for trichomes for which ancestral character state reconstructions based on binary characters (coded as presence/absence) suggested that the trait being considered was already present in the most recent common ancestor (MRCA) of the tribe. These analyses considered the presence of each trichome type in each plant part separately. As a result, ancestral state reconstructions of quantitative characters were not conducted for homoplasious trichome types. These analyses were performed exclusively with the Mk1 model with one parameter. In addition, ML reconstructions were used to assess the rates of transition of trichome traits in all ancestral character reconstructions.

RESULTS

Morphological characterization of trichome types

Two functional categories of trichomes were observed on vegetative plant parts of representatives of Bignonieae: (1) non-glandular and (2) glandular. Both trichome categories had variable sizes and cell numbers. All variants of non-glandular trichomes were included under the same category due to the similarities of these trichomes. On the other hand, three distinct morphotypes of glandular trichomes were recognized. These four trichome morphotypes (Fig. 2) were recorded on vegetative plant parts of Bignonieae species: (1) non-glandular trichomes (including unicellular simple, multicellular simple and multicellular branched – 'dentritic/stellar' – Fig. 2A–D); (2) peltate glandular trichomes (Fig. 2B, D, E–G); (3) stipitate glandular trichomes (Fig. 2H); and (4) patelliform/cupular glandular trichomes (Fig. 2I–K).

*Non-glandular (*Ng*) trichomes.* This trichome morphotype was present in all 105 species and 21 genera of Bignonieae sampled (Fig. 3). The distribution of this morphotype varied in terms of the position, density and plant size. These trichomes were uni- to multicellular, uniseriate, simple or branched, and formed by 1–25 cells, characterizing short and long trichomes. Trichome cells were covered by a thick, smooth or rough cuticule. Cuticular warts were present in some cases (e.g. *Anemopaegma*). Approximately 90 % of the species sampled (95/105) had simple trichomes with one to multiple cells (Fig. 3A–L) with variable position on the plants among species; the remaining 10 % of the species had branched trichomes with multiple cells (Fig. 3M–R) also with variable position on the plants among species. Multicellular and branched trichomes were encountered in some representatives of six different genera (i.e. *Adenocalymma*, *Amphilophium*, *Callichlamys*, *Fridericia*, *Manaosella* and *Pleonotoma*), not characterizing any particular genus. This trichome type occurred more widely in *Amphilophium* (Fig. 3N, O).

Although *Ng* trichomes were encountered in all sampled species, their presence was highly variable on the different plant parts among species (topological/positional variation). Only in 39 % of the species (41 out of 105) were these trichomes encountered on all vegetative parts, although different trichome densities were observed. In 7·6 % of the species (8/105), trichomes were distributed across all plant parts examined except leaflets. Overall, 82 % of the species (86/105) had trichomes on both sides of leaflets. In plants with trichomes on the abaxial side of leaflets, 19·3 % of the species (18/93) had trichomes growing exclusively over the leaflet veins. In the remaining 80·7 % of the species (75/93), trichomes were found across the abaxial side of the leaflet blade (Fig. 3A, K, M). In plants with trichomes on the adaxial side of leaflets, 44·2 % of the species (38/86) had trichomes growing exclusively over the leaflet veins (Figs. 3I, N). Approximately one-third of all species sampled (35/105) had *Ng* trichomes that were restricted to individual plant parts. In general, these species had trichomes that were restricted to the leaflet margin (e.g. *Periatomega vellozoi*) and/or to the ridges of the interpetiolar region (e.g. *Mansoa difficilis*), and/or to the apex of the prophylls of the axillary buds (e.g. *Lundia nitidula*). In addition, 27·6 % of the species (29/105) had high trichome densities (>10 trichomes mm^{-2}) on the abaxial surface of the leaflet blades. For species with *Ng* trichomes on both sides of the leaflets, the abaxial surface included approx. 16·5 times more trichomes per mm^2 on average than the abaxial side ($n = 63$).

Peltate glandular (Pg) trichomes. This trichome morphotype was present in all 105 species and 21 genera of Bignonieae sampled (Fig. 4), with variable density and size among plants. *Pg* were

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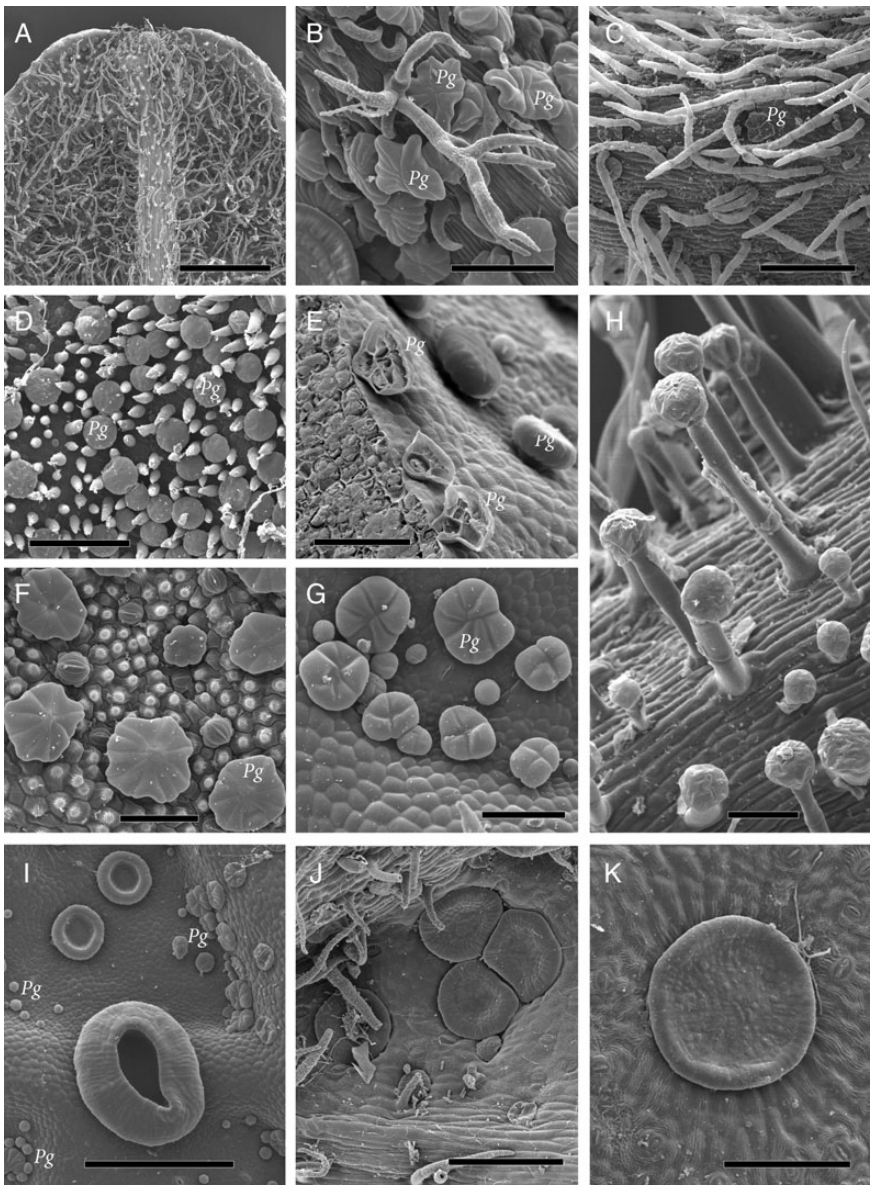


FIG. 2. SEM micrographs of trichome morphotypes distributed on vegetative plant parts of Bignonieae. (A–D) Non-glandular trichomes: (A) simple trichomes on the abaxial side of leaflets of *Anemopaegma scabriusculum*; (B) branched trichomes and peltate glandular trichomes on the leaflet vein of *Amphilophium crucigerum*; (C) simple trichomes on the leaflet vein of *Tanaecium pyramidatum*; (D) simple trichomes and peltate glandular trichomes (*Pg*) on the stems of *Periathomega vellozoi*. (E–G) Peltate glandular trichomes: (E) *Anemopaegma album*; (F) *Amphilophium crucigerum*; (G) *Adenocalymma pedunculatum*. (H) Stipitate glandular trichomes with adhesive secretion on the petiole of *Cuspidaria sceptrum*. (I) Patelliform/cupular glandular trichomes on the leaflets of *Adenocalymma pedunculatum* with one mature cupular trichome and two young trichomes (upper–central/left) on the abaxial side of leaflets. (J) *Anemopaegma album* with clusters of patelliform trichomes on the basal portion of the abaxial side of leaflets. (K) *Perianthomega vellozoi* with one patelliform trichome in the abaxial side of the leaflet blade. *Pg*, peltate glandular trichome. Scale bars: (A) = 1 mm; (B, D, K) = 100 μm; (C) = 150 μm; (E–H) = 50 μm; (I) = 250 μm; (J) = 200 μm.

multicellular, non-vascularized, stalked (a short stalk) and convex or rounded at earlier stages of development (sometimes becoming flattened with age). Each trichome had three distinct cell types. (1) Head cells: the head was composed of 6–24 thin-walled cells that were covered by a thin and smooth cuticle, but lacked visible pores. The trichome head also had a sub-cuticular space that accumulated secretions in many species (Fig. 4H2, K2). (2) Stalk cells: the stalk of trichomes was generally composed of a single cell in most species (Fig. 4F3, H3, K2) but included two or three cells in some taxa (e.g. *Stizophyllum*). These cells are rectangular (Fig. 4J4) or in the shape of an inverted trapezoid in most species (Fig. 4G3, H4). (iii) Foot cells. The foot was formed by two cells and generally is in the same plane as other epidermal cells. The *Pg* trichomes were similar among different species, with the number of head cells varying with development (Fig. 4F–M). Probably, in the post-secretory stage, the head of the *Pg* trichome disintegrates with cell death (more visible in Fig. 4G3, L3).

The *Pg* trichomes had a more homogeneous distribution across vegetative plant parts, and a higher variation in the density and size

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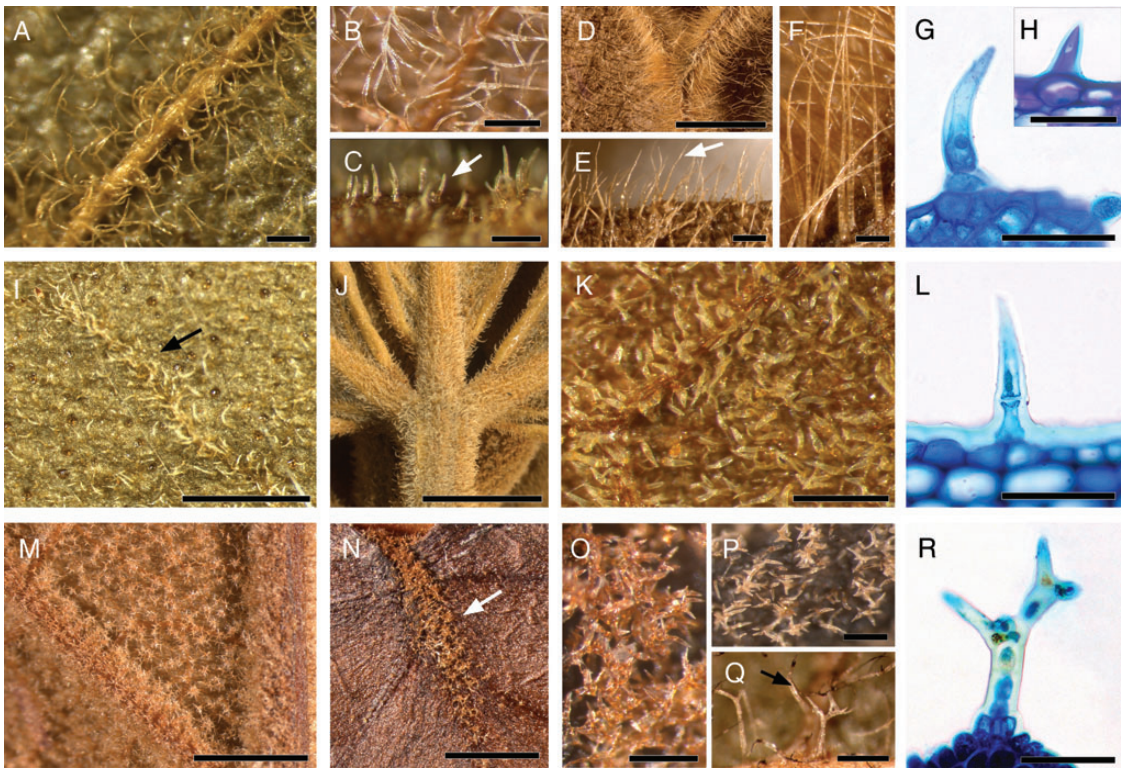


FIG. 3. Distribution and structure of non-glandular trichomes in Bignonieae. (A–L) Simple (unbranched) non-glandular trichomes on different plant parts. Unbranched trichomes were distributed on the abaxial blade of leaflets in *Anemopaegma setilobum* (A) and *Cuspidaria sceptrum* (B), and on the adaxial blade of *Adenocalymma pubescens* (C). Long trichomes of *Amphilophium askersonii* on petiole (D), leaflets (E) and stems (F). Arrows in C and E indicate one simple trichome. Photomicrograph of microtome sections of simple trichomes on the petiole of *Adenocalymma pedunculatum* (G); with two cells, and unicellular trichomes on the edge of leaflets of *Perianthomega vellozoi* (H). Simple trichomes on the leaflet vein and petiole + petiolule of *Pyrostegia venusta* (I) and *Anemopaegma scabriusculum* (J), respectively; and trichomes on the abaxial blade of leaflets in *Fridericia lasiantha* (K). Photomicrograph of a microtome section of simple multicellular trichomes of *Cuspidaria sceptrum* (L). (M–R) Branched non-glandular trichomes on different plant parts. Branched trichomes were distributed on the abaxial blade of leaflets in *Fridericia cinnamomea* (M), leaflet veins in *Amphilophium frutescens* (N, O), and with variable densities on the leaflets between *Fridericia nigrescens* (P) and *Fridericia dispar* (Q). Arrows in I and N indicate the leaflet vein; the arrow in Q indicates a branched trichome. Photomicrograph of microtome section of the branched multicellular trichomes on the leaflets of *Amphilophium parkerii* (R). Scale bars: (D, J) = 5 mm; (A–C, F, K, O, P, Q) = 0·25 mm; (G, H, L) = 50 μm; (M, N) = 2 mm; (I) = 1 mm; (E, R) = 0·5 mm.

of trichomes. The diameter of the glandular head varied between 0·02 and 0·10 mm^2 (median = 0·048, $n = 103$), with larger heads generally having more cells. Species of *Pyrostegia* and *Stizophyllum* had the largest glandular heads (up to 0·1 mm^2), and species of *Amphilophium* had the highest trichome densities (up to 216 trichomes mm^{-2} in *Amphilophium paniculatum*). Trichomes of *Pyrostegia*, *Amphilophium* and *Stizophyllum* (14/105 species sampled) reflected light when observed under the microscope (Fig. 4B, E). In some cases, peltate glandular trichomes were difficult to visualize without magnification lenses either because of their small size (glandular head diameter <0·028 mm; e.g. *Tynanthus*), or because trichomes were encountered in low densities (0·6 trichomes mm^{-2}; e.g. most species of *Adenocalymma*; Fig. 4A). In 42 % of the species sampled (44 out of 105), the leaflet blade had high or intermediate densities of *Pg* trichomes (>5 trichomes mm^{-2}) (Fig. 4B, C, E). Similar to the patterns found for the *Ng* trichomes, the abaxial side of leaflets had 5·3 times more *Pg* trichomes by mm^2 on average than the adaxial side ($n = 97$). Furthermore, *Pg* trichomes were generally more easily visualized on the abaxial side of the leaflet blades than on other vegetative plant parts (Fig. 4A–E). Trichomes were rarely observed on older portions of the interpetiolar regions, petioles and petiolules, suggesting that this trichome type is generally lost with age and/or secondary growth.

*Stipitate glandular (*Sg*) trichomes.* This trichome morphotype occurred only in nine of the 105 species sampled, and was encountered in six of the 21 genera of Bignonieae currently recognized (Fig. 5). In all species, the *Sg* trichomes had similar anatomical features that included a secretory head, and a multicellular, uniseriate, simple, non-vascularized stalk (generally long-stalked) with a variable number of cells among species. In general, the foot cells are at the same level and have the same morphology as other epidermal cells around the *Sg* trichome. (1) Head cells. A group of 8–12 thin-walled secretory cells. (2) Stalk cells. Include 3–12 cells, with the relative length of each stalk cells being greater than its width (2–3 times longer in mature trichomes), differentiating this trichome type from the *Pg* trichomes (Fig. 5E).

The *Sg* trichomes were rare and restricted to a few species. Plants with *Sg* trichomes showed high levels of variation in the distribution and density of trichomes among species (Fig. 5F–K). In some taxa, these trichomes were associated with the inflorescence (e.g. *Lundia densiflora* and *Manaosella cordifolia*). In others,

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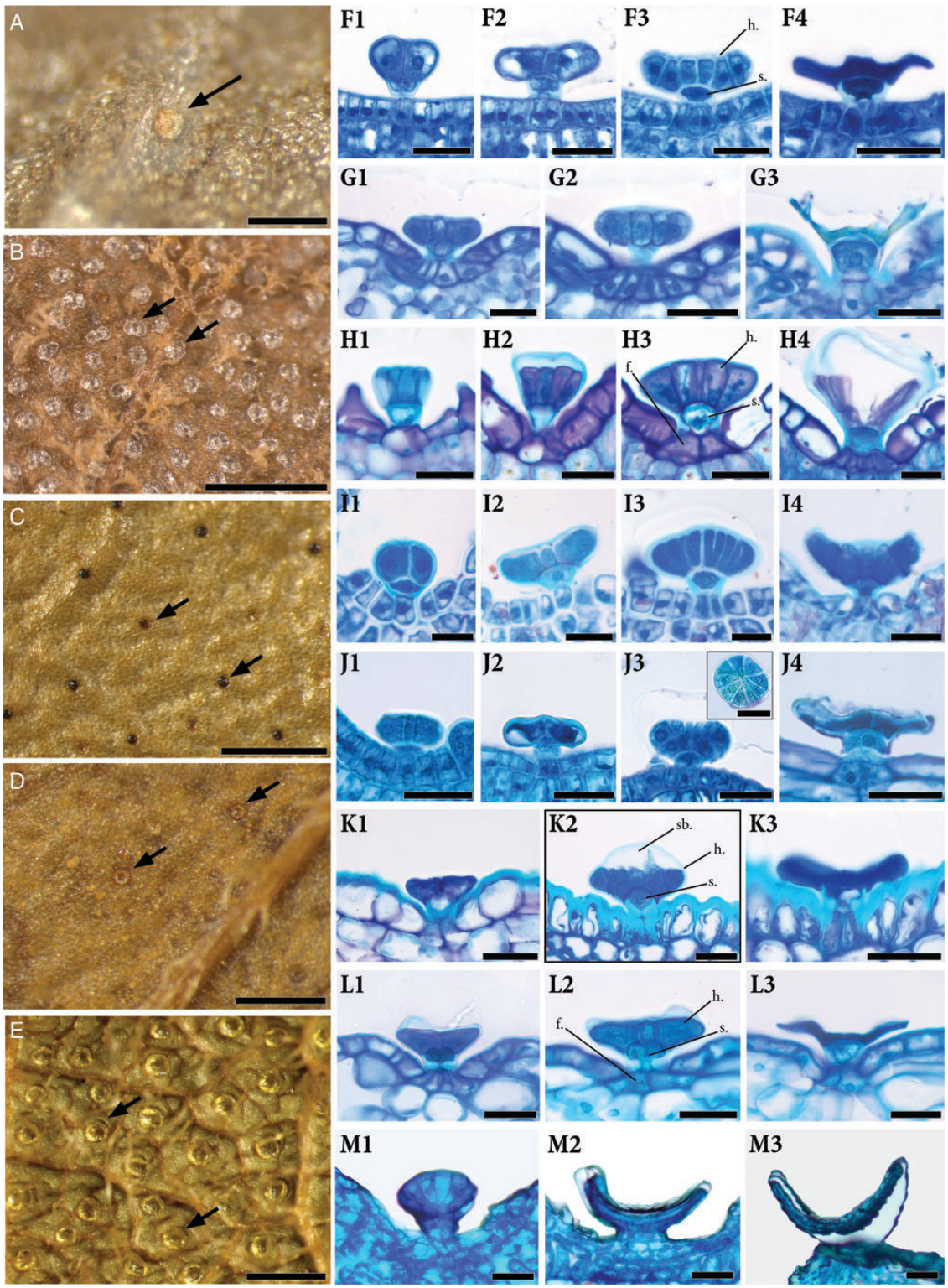


FIG. 4. Structure and development of peltate glandular trichomes in Bignonieae. (A–E) Traditional 'lepidote' indumentum in the abaxial side of leaflets showing the typical trichomes (arrows) of *Adenocalymma cymbalum* (A), *Amphilophium paniculatum* (B), *Anemopaegma chamberlaynii* (C), *Pleonotoma jasminifolia* (D), *Stizophyllum perforatum* (E). (F–M) Developmental stages of peltate glandular trichomes on leaflets: *Adenocalymma pedunculatum* (F), *Bignonia prieurei* (G), *Amphilophium crucigerum* (H), *Anemopaegma scabriusculum* (I1–I3) and *A. album* (I4), *Fridericia triplinervia* (J; inset in J3, frontal view), *Pleonotoma albiflora* (K), *Tanaecium pyramidatum* (L) and *Stizophyllum riparium* (M). Abbreviations: h., head cells; s., stalk cells; f., foot cells; and sb., sub-cuticular space with secretion. Scale bars: (A, D) = 0·25 mm; (B, C, E) = 0·50 mm; (F–M) = 25 μm.

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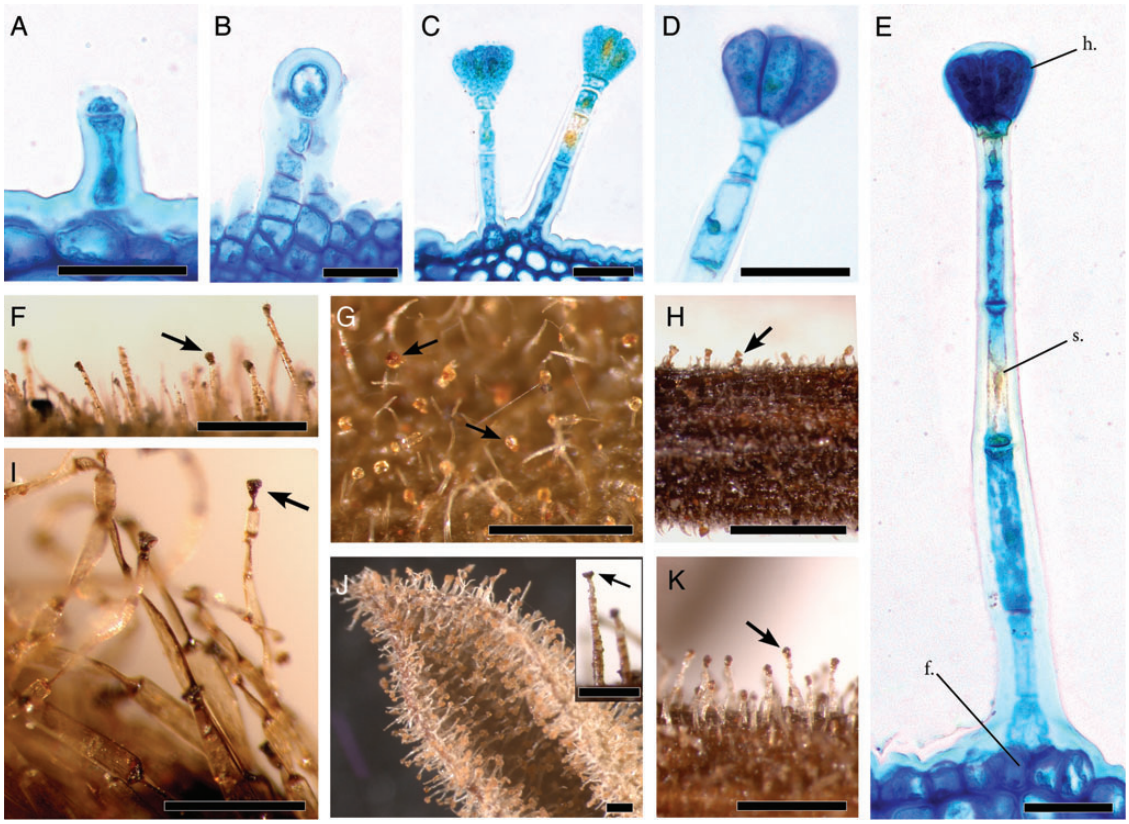


FIG. 5. Structure and development of stipitate glandular trichomes in Bignonieae (arrows). (A–E) Developmental stages of stipitate glandular trichomes of *Cuspidaria sceptrum*. (F–K) Glandular trichomes on the petiole of *Adenocalymma adenophorum* (F), leaflets of *Fridericia erubescens* (G), petioles of *Manaosella cordifolia* (H), stems of *Adenocalymma trichocladum* (I), in young leaves near the inflorescence of *Mansoa hirsuta* (J) and in petioles of *Martinella obovata* (K). Abbreviations: h., head cells; s., stalk cells; f., foot cells; Scale bars: (A–E) = 50 μm; (F, G, I, K) = 0·5 mm; (H, J) = 0·25 mm.

trichomes were dispersed throughout the plant, independently of the inflorescence position (e.g. *Adenocalymma adenophorum*). *Cuspidaria sceptrum* (dry habitat) and *A. adenophorum* (forest habitat) produced a sticky secretion that detained insects (e.g. Diptera). However, no ants were observed visiting these plants (A. Nogueira, pers. obs.). Developmental analyses of the *Sg* trichomes (i.e. *C. sceptrum*; Fig. 5A–E) and *Pg* trichome (Figs. 4F–M) corroborated the categorization of these trichomes into two distinct morphotypes.

*Patelliform/cupular glandular (*PCg*) trichomes.* This trichome morphotype was present in all 105 species and 21 genera of Bignonieae (Fig. 6), distributed in different positions, with variable abundance and size among plants. The *P/Cg* trichomes were multicellular, non-vascularized, disc-shaped (patelliform) or cup-shaped (cupular), with a concave surface, and three distinct cell layers. (1) Secretory layer. This layer was formed by 30–68 thin-walled columnar cells that were covered by a thin and smooth cuticle (Fig. 6F, P), but lacked visible pores. The secretory cells were organized in a palisade-like arrangement and had a densely staining cytoplasm. (2) Intermediate layer. This layer had two different morphologies depending on the taxa. Some taxa had an intermediate layer composed of large, ellipsoid, vacuolated cells, surrounded by thick and highly suberized anticlinal walls (Fig. 6C, D, F, I, J, L, M), whereas others had an intermediate layer that was composed of one row of thick, highly suberized cells that connected the secretory cells to the epidermis (Fig. 6G, O, P). (3) Foot or basal layer. This layer was multicellular, and consisted of small quadrangular cells.

Overall, *P/Cg* trichomes are flattened structures (Fig. 6E, F, H–K, M, N–P). Although most species of Bignonieae had flattened trichomes, most species of *Adenocalymma* had cupular trichomes, the cupular shape being caused by cell proliferation and expansion in the upper and central direction at the periphery of the secretory layer, creating a pocket-like structure (Fig. 6A, D, I). In some cases (e.g. *A. cymbalum*), the expansion of the surrounding epidermis was very pronounced, leading to a 'volcano-like' structure (*sensu* Lohmann, 2006). A few species (e.g. *Amphilophium parkerii*, *Bignonia prieurei*) had trichomes with morphology intermediate between the cupular and patelliform variants (Fig. 6G, J). These intermediate morphologies were also observed in a single species (e.g. *Adenocalymma pedunculatum*, Fig. 6C).

The abundance of *P/Cg* trichomes was highly variable among different plant parts, varying from sparsely distributed in some species to highly clustered in others. Although all 105 species sampled had *P/Cg* trichomes, 24·8 % of the species sampled (26/105) had a few *P/Cg* trichomes scattered on the leaflet blades (e.g. *Adenocalymma impressum*, *Manaosella cordifolia*, *Periathomega vellozoi*, *Pyrostegia venusta* and *Tanaecium pyramidatum*). In contrast, 79 of the species sampled had clusters of *P/Cg* trichomes (>5 trichomes mm^{-2}) in different plant parts

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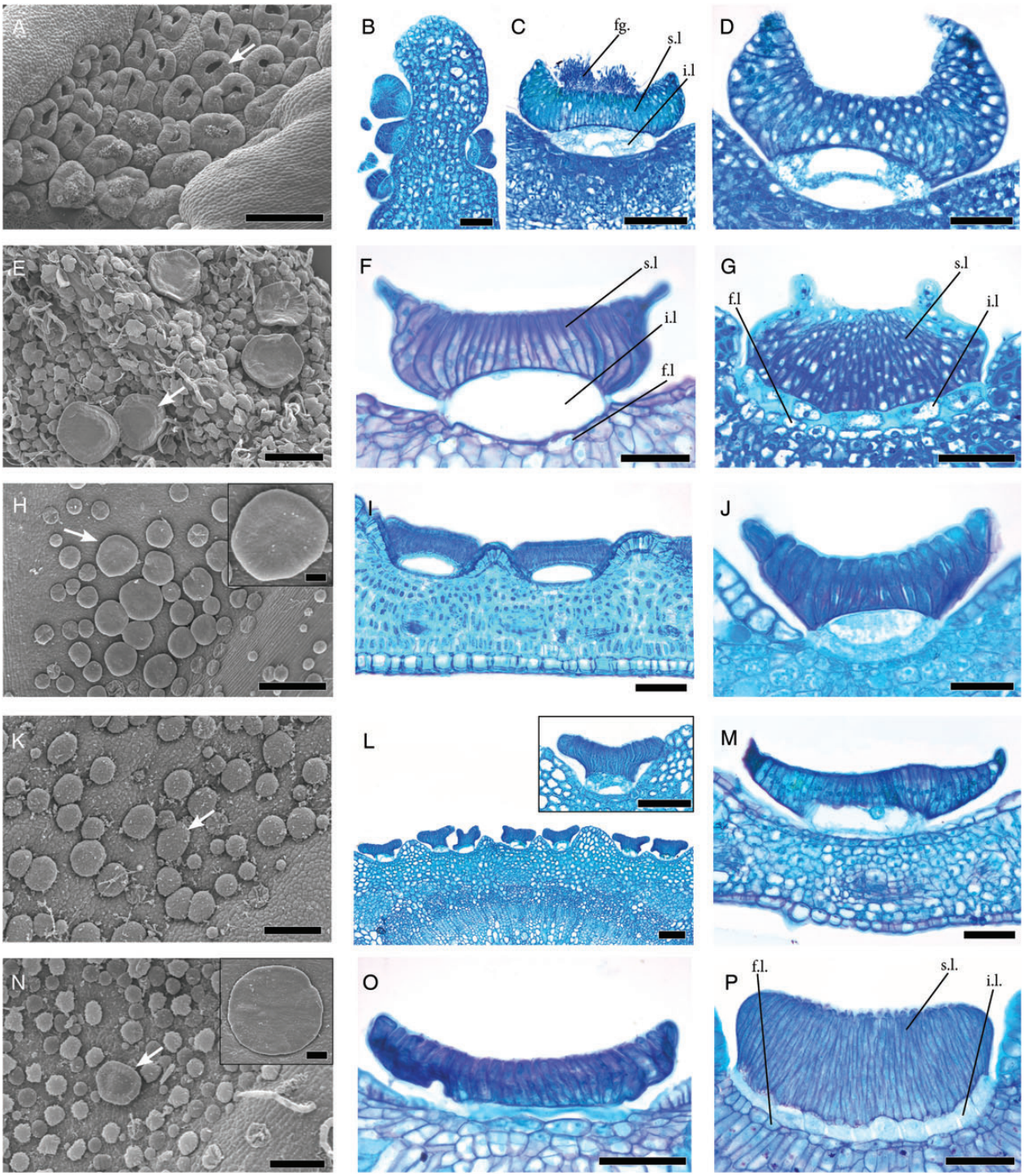


FIG. 6. Morphoanatomical variation of patelliform/cupular glandular trichomes in Bignonieae. (A–D) Patelliform/cupular trichomes of *Adenocalymma peduncu-latum*: (A) SEM micrograph of mature prophylls with aggregated cupular trichomes; (B, C) photomicrograph of microtome section of young prophylls with developing trichomes (B), trichomes infected by fungi (C) and cupular trichomes on leaflets (D). (E–G) Patelliform glandular trichomes in *Amphilophium*: (E) SEM micrograph of abaxial side of young leaflets of *A. crucigerum*; (F, G) photomicrograph of microtome section of patelliform trichomes on the leaflets of *A. crucigerum* (F) and *A. parkerii* (G). (H) SEM photomicrographs of aggregated patelliform trichomes at the base of the abaxial side of leaflets of *Mansoa difficilis* (inset: detail of individual trichome). (I, J) Photomicrograph of microtome section of patelliform trichomes at the base of young leaflets (abaxial side) of *Anemopaegma album* (I) and on the prophylls (adaxial side) of *Bignonia prieurei* (J). (K, L) Aggregated trichomes on the young stems (interpetiolar region) of *Lundia nitidula* (K; SEM photomicrograph) and *Fridericia triplinervia* (L; microtome sections; inset: detail of individual trichome). (M) Patelliform trichome on the abaxial side of the leaflets of *Cuspidaria sceptrum* (microtome section). (N) SEM photomicrographs of trichomes on the abaxial side of the leaflets of *Tanaecium pyramidatum* (inset: detail of individual trichome). (O, P) Photomicrograph of microtome section of trichomes on the adaxial side of the leaflets (O) and prophylls of *Pleonotoma albiflora* (P). Arrows in A, E, H, K, N indicate each trichome type, where these are most evident. Abbreviations: s.l., secretory layer; i.l., intermediate layer; f.l., foot or basal layer; and fg., fungi growing on the trichome secretory tissue. Scale bars: (A) = 350 μm; (B, I, L, N) = 100 μm; (D, F, G, J, M, O, P) = 50 μm; (E) = 200 μm; (H) = 150 μm, inset = 50 μm; (K) = 110 μm; (N) = 60 μm (inset).

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(Fig. 6A, H, K, L). In particular, 37·1 % of the species (39 out of 105) had clusters of *P/Cg* trichomes at the base of leaflets (abaxial side); this feature was particularly common in representatives of *Mansoa* (Figs 6H, I). Furthermore, 35·2 % of the species (37 out of 105) had clusters of *P/Cg* trichomes at the interpetiolar region of stems; this feature was particularly common in representatives of *Lundia* and *Fridericia* (Fig. 6L). In addition, 22·8 % of the species sampled (24 out of 105) had clusters of *P/Cg* trichomes on the prophylls of the axillary buds; this feature was particularly common in representatives of *Adenocalymma*, *Bignonia* and *Pleonotoma* (Fig. 6A, B). Clusters of *P/Cg* trichomes were rarely documented on petioles and petiolules, found only in 11/105 species sampled; clustered *P/Cg* trichomes were particularly common in representatives of *Pachyptera* and in the monotypic *Callichlamys*. *P/Cg* trichomes clustered on the adaxial side of the leaflet blades were rare and restricted to *Bignonia corymbosa*. In the field, *P/Cg* trichomes are yellowish-green and turgid when active, but brownish and dry when inactive. Active trichomes secreted hyaline and viscous nectar, which accumulated in the concave surface of trichomes and was eaten by insects, mostly ants.

Ancestral character state reconstructions

Initially we used information of positional and structural similarity to generate hypotheses of common origin. Four similar structural trichome types were described (see details before) on different vegetative plant parts. Thus, ML optimizations of the four trichome morphotypes (i.e. *Ng*, *Pg*, *Sg* and *P/Cg*) were performed on each plant part separately (i.e. interpetiolar region of stems, prophylls of the axillary buds, petiole, petiolule, and adaxial and abaxial surface of leaflets) (Table 1). Further details on each reconstruction are presented below.

*Non-glandular (*Ng*) trichomes.* The ML ancestral character state reconstrutions of *Ng* trichomes indicate that this structure was already present in the MRCA of Bignonieae in almost all vegetative plant parts (Table 1, Fig. 7A). In particular, ancestral state reconstructions indicated that *Ng* trichomes were already present on the stems (82 %), prophylls (99 %), petioles (99 %), petiolules (99 %) and leaflet veins (adaxial side with 95 %, and abaxial side with 81 %) in the MRCA of Bignonieae. On the leaflet vein of the abaxial side, the *Ng* trichome seems to have been lost at least once, and gained at least seven times (Fig. 7A). Furthermore, the MRCA of Bignonieae probably did not bear *Ng* trichomes on the adaxial (31 %) or abaxial (30 %) side of leaflets, with this trait appearing on the blade of the adaxial side of the leaflets at least 20 times, and on the blade of the abaxial side at least 16 times (Fig. 7A). When the two morphological variants of *Ng* trichomes were considered (unbranched and branched), ML reconstructions showed eight independent origins of the unbranched to branched trichomes (Fig. 7B). Ancestral character state reconstructions of trichome density and size on the abaxial blade of leaflets suggested that the MRCA of Bignonieae had a low density (40 %) of intermediate trichome size (37 %) (Table 2).

*Peltate and glandular (*Pg*) trichomes.* Ancestral character state reconstructions of *Pg* trichomes indicate that the MRCA of Bignonieae already had *Pg* trichomes in all plant parts (Table 1, Supplementary Data Fig. S1), the most widespread type of trichome in Bignonieae. In particular, this trichome type was probably present on the stems (87 %), prophylls (99 %), petioles (93 %), petiolules (90 %), and adaxial (79 %) and abaxial (99 %) side of leaflets in the MRCA of Bignonieae. Even though *Pg* trichomes were probably present in the MRCA of Bignonieae on the adaxial side of leaflets, these trichomes seem to have been lost at least once in the ancestor of the clade *Adenocalymma* + *Neojobertia*, and re-gained five times in this clade. Ancestral character state reconstructions of quantitative features of *Pg* trichomes suggested that the MRCA of Bignonieae probably had a high density (35 %) and intermediate size (38 %) on the abaxial side of leaflets (Table 2, Fig. 8). The largest and densest peltate glandular trichomes were observed in *Amphilophium*, *Pyrostegia* and *Stizophyllum*.

*Stipitate glandular (*Sg*) trichomes.* The MRCA of Bignonieae did not bear stipitate glandular trichomes on all vegetative plant parts sampled (Table 1, Fig. 9). Nine independent gains of *Sg* trichomes were hypothesized in Bignonieae (Fig. 9), in the following genera: *Adenocalymma* (two gains), *Fridericia* (one gain), *Xylophragma* (one gain), *Cuspidaria* (two gains), *Lundia* (one gain), *Manaosella* (one gain) and *Martinella* (one gain). The morphological differences observed in *Sg* among taxa and the lack of a historical connection between those trichomes in the various lineages suggest that the various *Sg* trichomes encountered in Bignonieae did not have the same origin in this plant group. Thus, the evolution of quantitative features related to these trichomes was not explored further.

*Patelliform/cupular glandular (*PCg*) trichomes.* The MRCA of Bignonieae probably only had *P/Cg* trichomes on the abaxial side of leaflets (Table 1, Fig. 10). In particular, this trichome type was probably not present on stems (35 %), prophylls (3 %), petioles (5 %), petiolules (1 %) or the adaxial side of leaflets (78 %) of the MRCA of Bignonieae. In addition, *P/Cg* trichomes probably evolved multiple times in each plant part (Fig. 10). More specifically, at least six independent evolutionary events of *P/Cg* trichomes were hypothesized for the interpetiolar region of stems of Bignonieae, including an origin in the MRCA of the '*Tanaecium* + *Fridericia* + *Xylophragma* + *Cuspidaria* + *Tynanthus* + *Lundia*' clade, an origin in the MRCA of *Dolichandra* and an origin in the MRCA of *Martinella*. Similarly, at least six origins of *P/Cg* trichomes were hypothesized for the prophylls of the axillary buds, including one in the MRCA of *Adenocalymma*, one in the MRCA of *Pleonotoma* and one in the MRCA of the '*Amphilophium* + *Anemopaegma* + *Pyrostegia* + *Mansoa* + *Bignonia*' clade. Ancestral state reconstructions of quantitative features of *P/Cg* trichomes suggested that the MRCA of Bignonieae showed a low density of *P/Cg* trichomes on the adaxial side of leaflets (49 %), a high density on the abaxial side (41 %) and small glandular heads (34 %) (Table 2).

DISCUSSION

In this study we encountered four main trichome morphotypes (i.e. non-glandular, peltate glandular, stipitate glandular and patelliform/cuputar glandular) with a highly variable position on the vegetative plant parts (i.e. stems, prophylls, petiole, petiolule, and adaxial and abaxial blade of leaflets) among species of Bignonieae. Three of the four trichome morphotypes (*Ng*, *Pg* and

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TABLE 1. *ML ancestral character state reconstructions of different trichome types in the MRCA of tribe Bignonieae (Bignoniaceae)*

Trichome characters			Two-rate model		One-rate model			Estimated ancestral character states for the tribe Bignonieae (proportional likelihood of trichome presence)	
Types		Plant parts	Rate of evolution (gain, loss of trichomes)	Likelihood	Estimated rate of evolution	Likelihood	Was a two-rate model favoured?		
1	*Ng*	Interpetiolar regions (stems)	**0·124, 0·026**	**47·85**	0·011	51·30	**Yes**	Presence (0·82)	
2	*Ng*	Prophylls	0·007, 0·00005	5·07	**0·00005**	**5·63**	No	Presence (0·99)	X
3	*Ng*	Petioles	0·051, 0·004	28·98	**0·005**	**30·53**	No	Presence (0·99)	X
4	*Ng*	Petiolules	0·067, 0·009	34·89	**0·006**	**36·43**	No	Presence (0·99)	X
5	*Ng*	Leaflet vein (adaxial side)	0·059, 0·016	49·08	**0·011**	**50·33**	No	Presence (0·95)	
6	*Ng*	Leaflet blade (adaxial side)	0·022, 0·035	70·28	**0·029**	**71·32**	No	Absence (0·31)*	
7	*Ng*	Leaflet vein (abaxial side)	0·027, 0·011	54·03	**0·014**	**55·40**	No	Presence (0·81)*	
8	*Ng*	Leaflet blade (abaxial side)	0·021, 0·022	67·53	**0·022**	**67·53**	No	Absence (0·30)*	
9	*Pg*	Interpetiolar regions (stems)	**0·058, 0·005**	**29·06**	0·005	33·53	**Yes**	Presence (0·87)	
10	*Pg*	Prophylls	0·536, 0·006	5·56	**5·607**	**7·21**	No	Presence (0·99)	X
11	*Pg*	Petioles	**0·041, 0·003**	**20·41**	0·003	23·26	**Yes**	Presence (0·93)	X
12	*Pg*	Petiolules	**0·044, 0·003**	**22·81**	0·003	26·91	**Yes**	Presence (0·90)	X
13	*Pg*	Leaflet blade (adaxial side)	**0·035, 0·001**	**18·42**	0·003	23·08	**Yes**	Presence (0·79)	
14	*Pg*	Leaflet blade (abaxial side)	0·060, 0·001	11·44	**0·001**	**12·88**	No	Presence (0·99)	X
15	*Sg*	Interpetiolar regions (stems)	**0·026, 0·273**	**30·69**	0·005	34·76	**Yes**	Absence (0·09)*	X
16	*Sg*	Petioles	**0·026, 0·273**	**30·69**	0·005	34·76	**Yes**	Absence (0·09)*	X
17	*Sg*	Petiolules	**0·026, 0·273**	**30·69**	0·005	34·76	**Yes**	Absence (0·09)*	X
18	*Sg*	Leaflet blade (adaxial side)	**0·024, 0·290**	**28·27**	0·004	31·90	**Yes**	Absence (0·08)*	X
19	*Sg*	Leaflet blade (abaxial side)	**0·020, 0·240**	**28·26**	0·004	31·55	**Yes**	Absence (0·08)*	X
20	*P/Cg*	Interpetiolar regions (stems)	**0·006, 0·018**	**48·55**	0·011	50·72	Yes	Absence (0·35)*	
21	*P/Cg*	Prophylls	0·005, 0·014	43·54	**0·008**	**45·39**	No	Absence (0·03)*	X
22	*P/Cg*	Petioles	0·014, 0·032	52·81	**0·013**	**53·15**	No	Absence (0·05)*	X
23	*P/Cg*	Petiolules	0·003, 0·027	20·69	**0·003**	**21·21**	No	Absence (0·01)*	X
24	*P/Cg*	Leaflet blade (adaxial side)	**0·043, 0·009**	**44·95**	0·009	47·48	**Yes**	Presence (0·78)*	
25	*P/Cg*	Leaflet blade (abaxial side)	**0·481, 0·009**	**9·86**	0·001	12·18	**Yes**	Presence (0·98)	X

Likelihoods are reported as negative logarithms and the ancestral character state is presented in the last column. Estimates of the rates of evolution are also presented.
Ng, non-glandular trichomes; *Pg*, peltate glandular trichomes; *Sg*, stipitate glandular trichomes; *P/Cg*, patelliform/cupular glandular trichomes.
*Multiple independent gains in Bignonieae.
X = most likely states according to a decision threshold T.
The estimated parameters of the evolutionary model favoured are presented in bold.

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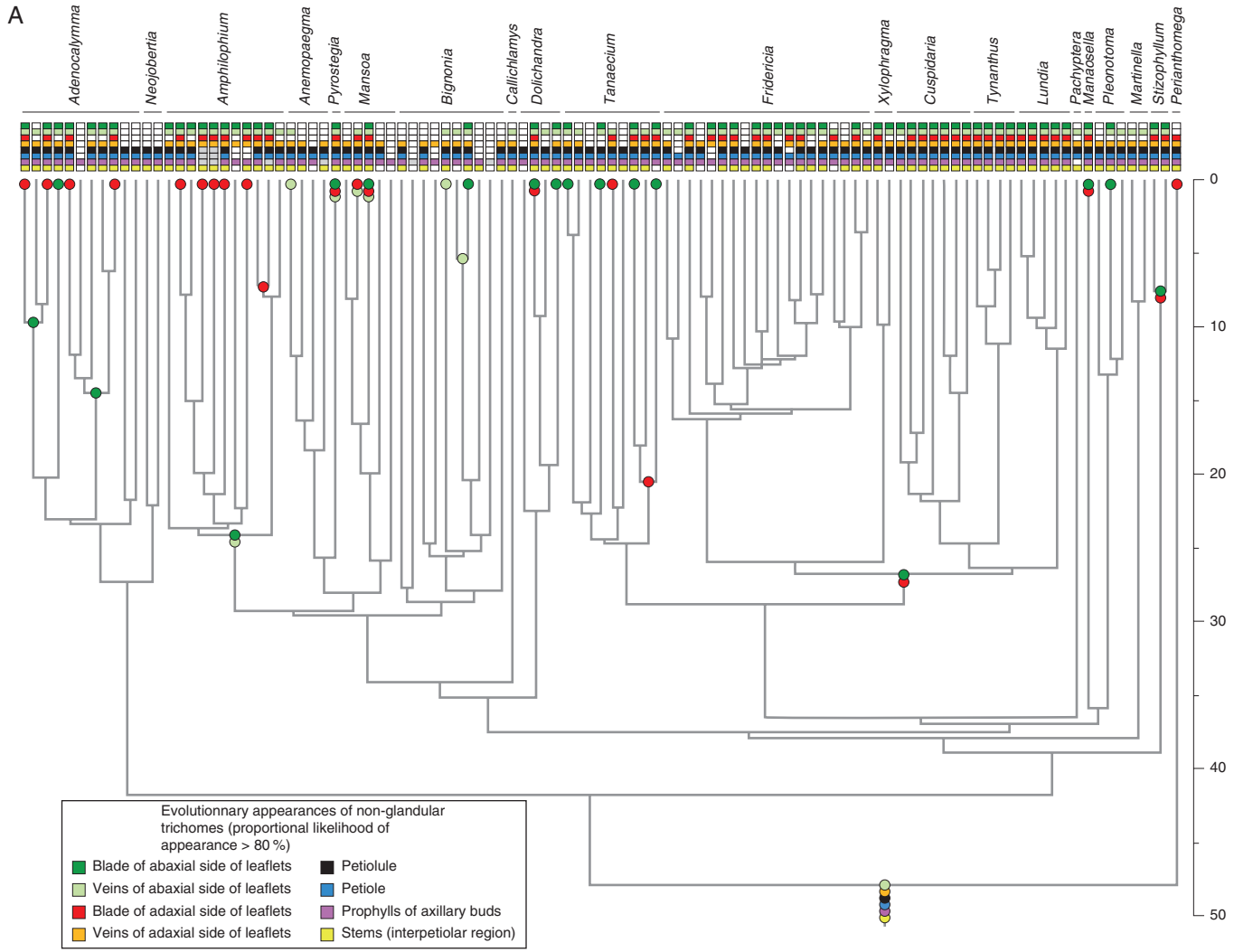


FIG. 7. Evolution of non-glandular trichomes. (A) The evolutionary appearance of trichomes (proportional likelihood of appearance >80 %) is shown by rectangles with different colours, according to the appearance of non-glandular trichomes on different vegetative plant portions. Losses of these characters and the detailed maximum likelihood (ML) ancestral character state reconstructions are presented in Supplementary Data Fig. S1. (B) ML ancestral character reconstruction of non-glandular trichomes coded as 'simple' and 'dendritic' trichomes. Black portions of each pie chart represent the proportional likelihood of the appearance of dendritic non-glandular trichomes. The scale on the right is millions of years ago.

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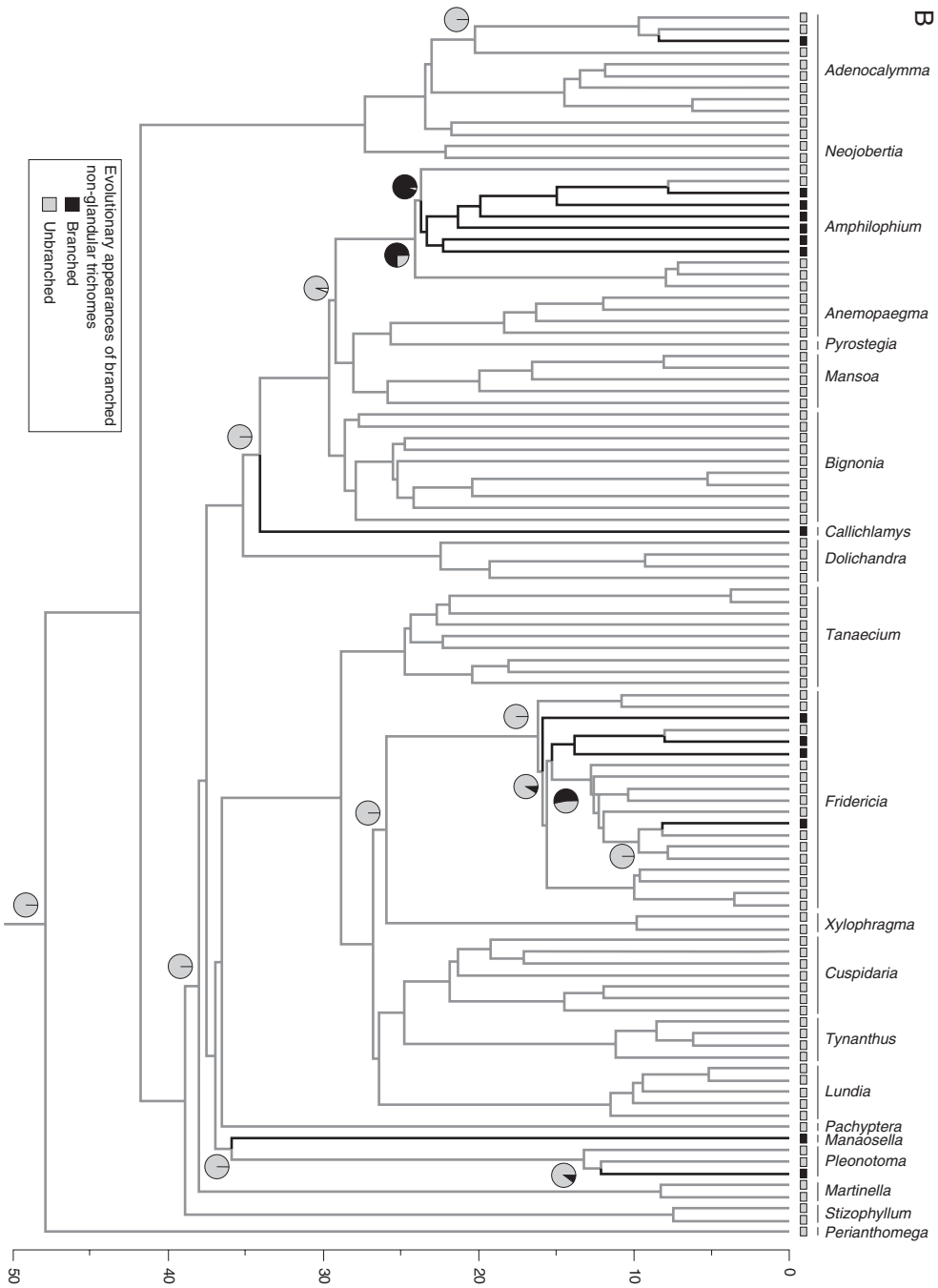


FIG. 7. (Continued)

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TABLE 2. *ML ancestral character state reconstructions of trichome quantitative traits (size and density/abundance) in the MRCA of tribe Bignonieae (Bignoniaceae)*

Trichome characters			One-rate model		
Types		Quantitative traits (and position on the plants)	Estimated rate of evolution	Likelihood	Estimated ancestral character state of tribe Bignonieae (Proportional likelihood)
1	*Ng*	Density (adaxial side of leaflets)	0·020	107·19	Small (0·40)
2	*Ng*	Size (adaxial side of leaflets)	0·031	80·32	Intermediate (0·37)
4	*Pg*	Density (abaxial side of leaflets)	0·036	109·20	High (0·35)
5	*Pg*	Size of glandular head (abaxial side of leaflets)	0·023	106·35	Intermediate (0·38)
6	*P/Cg*	Number of EFNs (adaxial side of leaflets)	0·024	110·67	Small (0·49)
7	*P/Cg*	Number of EFNs (abaxial side of leaflets)	0·022	109·02	Intermediate (0·41)
8	*P/Cg*	Size of the glandular head (abaxial side of leaflets)	0·031	110·73	Intermediate (0·34)

Likelihoods are reported as negative logarithms and the ancestral character state is presented in the last column, based on a one-rate model of evolution (Mk1). Estimates of the rates of evolution are also presented.
Ng, non-glandular trichomes; *Pg*, peltate glandular trichomes; *P/Cg*, patelliform/cupular glandular trichomes.
The size of the glandular head is the diameter of the glandular head.

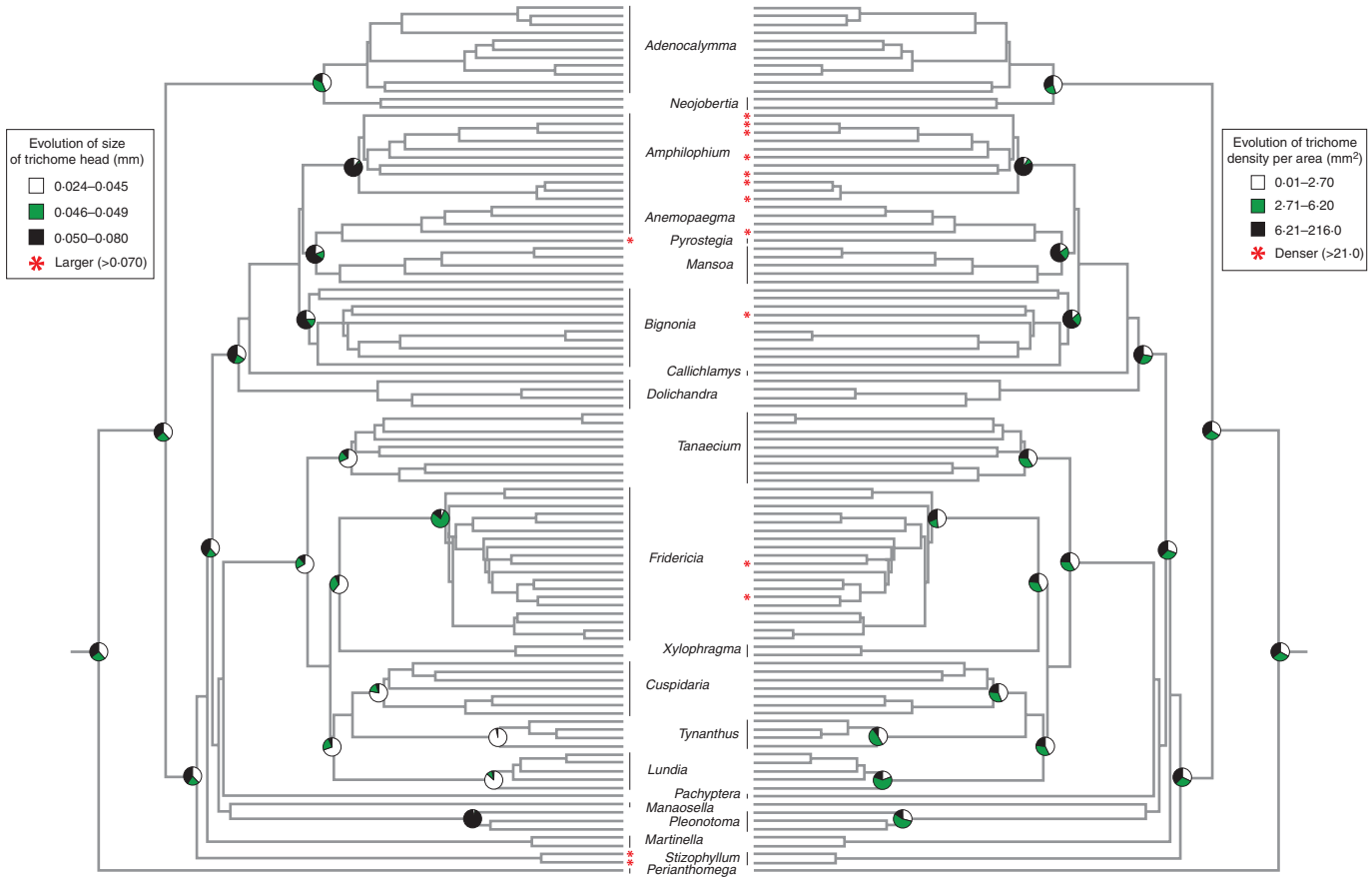


FIG. 8. Evolution of peltate glandular trichomes. Right: ML ancestral state reconstructions of trichome density. Left: ML ancestral state reconstructions of the diameter of the trichome glandular heads. Asterisks indicate the lineages in which peltate glandular trichomes are 'larger' and 'denser' in the tribe.

P/Cg) were already present in the MRCA of Bignonieae, but in general these trichome types only occur on some plant parts, with secondary occupations of each trichome type on new plant parts during the history of diversification of Bignonieae. At least for some plant parts, we corroborate the hypothesis of a single evolutionary origin for the different trichome types (e.g. *Ng*, *Pg* and *P/Cg* trichomes), suggesting that these structures are best treated under the same name based on their morphological similarity and common evolutionary history of such structures.

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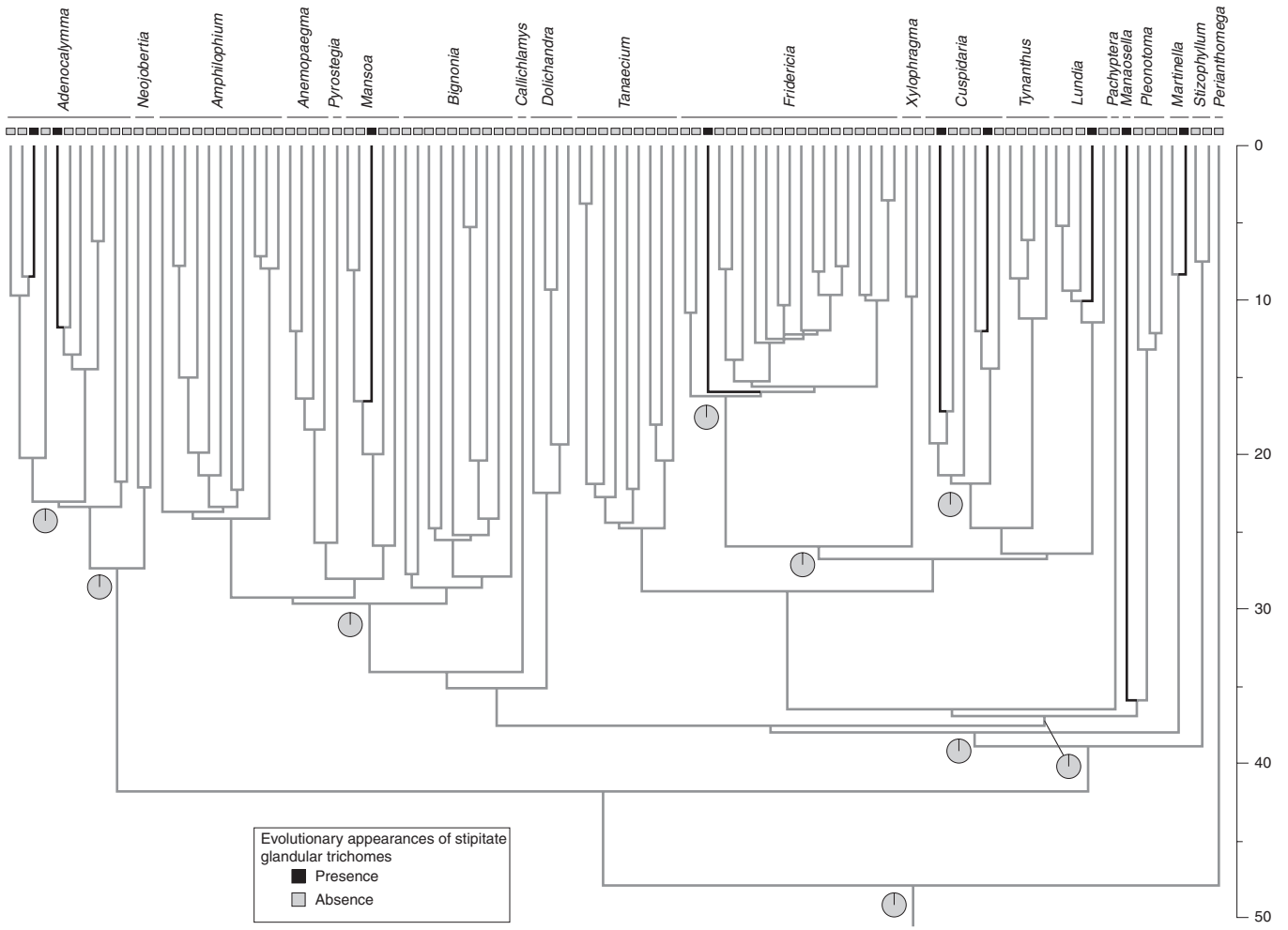


FIG. 9. Evolution of stipitate glandular trichomes. ML ancestral character state reconstruction indicated nine independent evolutions of this trichome morphotype in that lineage. Black portions of each pie chart represent the proportional likelihood of stipitate glandular trichome appearances. The scale on the right is millions of years ago.

Integrating morphological data and ancestral character state reconstructions

Trichomes often demonstrate variable macro- and micromorphological features, making it difficult to determine the exact trichome type that is being referred to in the literature (Theobald *et al.*, 1979). In the specific case of Bignoniaceae, a wide range of terms has been used to describe the same trichome type (Fig. 1). Below we summarize major morphoanatomical features and the evolutionary history of each trichome type.

*Non-glandular (*Ng*) trichomes.* The *Ng* trichomes of representatives of Bignonieae have only rarely been described in detail (e.g. Ogundipe and Wujek, 2004). Instead of being described in terms of their structure, *Ng* trichomes have been traditionally described in terms of their density and overall appearance. For example, 'villose leaves' have been described for *Anemopaegma velutinum*, referring to a hairy cover of dense, straight, long and soft trichomes (Bureau and Schumann, 1897). However, no information is available on the anatomy and developmental sequence of *Ng* trichomes in Bignoniaceae or other plant groups (e.g. Lamiaceae; Naidu and Shah, 1981). Even though *Ng* trichomes have generally been treated as morphologically homogeneous structures, the variable number of cells (and size) and variable patterns of distribution of this trichome morphotype on different plant parts indicate the need for more detailed morphoevolutionary studies of trichomes in general.

Ancestral state reconstructions of *Ng* trichomes in Bignonieae indicated that simple and unbranched trichomes were present in the ancestor of Bignonieae, with branching trichomes having evolved at least eight times. This transition from simple to branched trichomes was similar to that observed in Brassicaceae, in which the pattern of trichome evolution indicated numerous innovations of trichome branching (Beilstein *et al.*, 2006). Furthermore, a hierarchical pattern of occupation of trichomes over different vegetative plant parts was also observed during the evolutionary history of Bignonieae. More specifically, the MRCA of the tribe probably already had *Ng* trichomes on the branches, prophylls of the axillary buds, and petioles. However, it was only later that trichomes covered the surfaces of leaflet blades. In most species, the density of trichomes remained low and *Ng* trichomes remained unbranched. A few lineages recently acquired higher trichome densities (e.g. species of *Amphilophium* and *Tynanthus*) and/or have acquired branched trichomes (e.g. species of *Amphilophium* and *Fridericia*).

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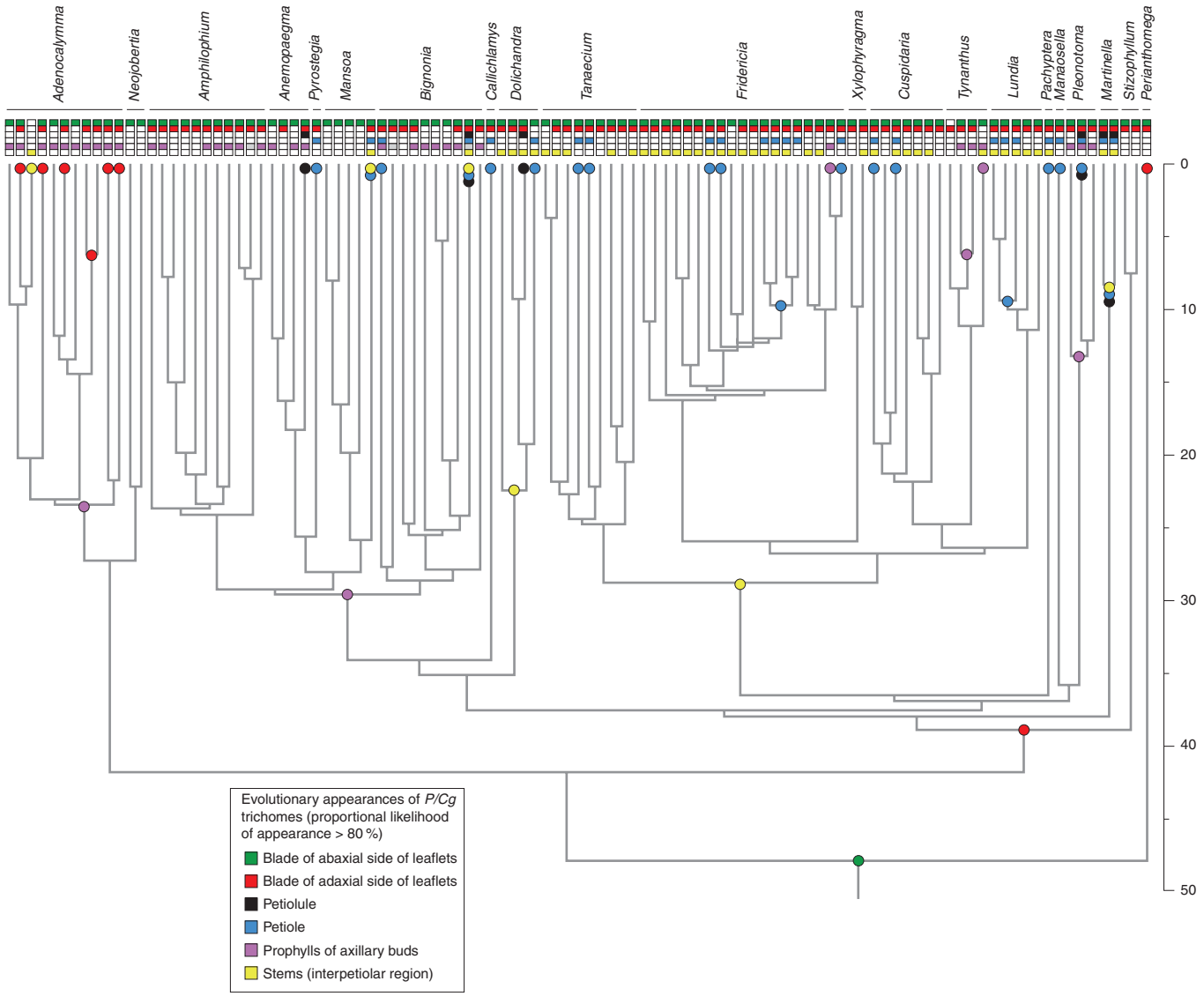


FIG. 10. Evolution of the patelliform/cupular glandular trichomes (*P/Cg*). The evolutionary appearance of trichomes (proportional likelihood of appearance >80 %) is shown by rectangles with different colours, according to the appearance of *P/Cg* trichomes on different vegetative plant portions. Losses of these characters and the detailed ML ancestral character state reconstructions are presented in Supplementary Data Fig. S1. The scale on the right is millions of years ago.

*Peltate glandular (*Pg*) trichomes.* The *Pg* trichomes were first recorded in representatives of Bignoniaceae in the 19th century in taxonomic treatments of the *Prodromus Systematis Naturalis Regni Vegetabilis* (Candolle, 1845) and *Flora Brasiliensis* (Bureau and Schumann, 1897). For example, in the *Flora Brasiliensis* (Bureau and Schumann, 1897), *Pg* trichomes were documented in *Cydista aequinoctialis* ('densius lepidota', p. 32), *Pithecoctenium dolichoides* ('conspicue lepidota obsolete pellucide punctulata', p. 22), *Pyrostegia venusta* ('lepidibus minutis inspersa et ope eorum plus minus manifeste pellucide punctate', p. 23) and *Stizophyllum inaequilaterum* ('pellucide punctate', p. 29). Seibert (1948) used two distinct terms to refer to the variants of *Pg* trichomes: 'pellucid glands' and 'glandular scales'. This author considered *Amphilophium*, *Pithecoctenium* (= *Amphilophium sensu* Lohmann, 2003; Lohmann and Taylor, 2013), *Pyrostegia* and *Stizophyllum* to have 'pellucid glands', and compared these structures with the translucent pellucid dots in leaves of species of Rutaceae. However, in Rutaceae, such structures are present in almost all members of the family and represent secretory cavities (Engler, 1931; Kubitzki *et al.*, 2011) that are immersed in the mesophyll, and not epidermal projections as in Bignoniaceae.

Seibert (1948) considered the majority of the representatives of Bignoniaceae to have 'glandular scales' (and a few 'pellucid glands') that he described as 'minute scales found on stems, petioles, leaves, calyx, corolla, ovary and fruit, and responsible for the 'lepidote' condition frequently encountered in the Bignoniaceae'. Gentry (1980), on the other hand, preferred to call these structures 'glandular-lepidote trichomes' instead of 'glandular scales'. The study of *Pg* trichomes conducted here revealed that even though these trichomes are identical in structure in all studied species, they are much denser in *Amphilophium*. All these structures are composed of a glandular convex or rounded head of a variable number of cells. Furthermore, this trichome type has a single origin in Bignonieae.

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The morphological similarity of *Pg* among species and the phylogenetic pattern indicated that this secretory structure is best treated uniformly under the same terminology. We here propose that these structures are best treated morphologically as peltate glandular trichomes.

Ancestral character state reconstructions indicate that *Pg* trichomes were already present on both sides of the leaflet blades and all other vegetative portions of the MRCA of Bignonieae. However, *Pg* trichomes probably originated prior to the origin of Bignonieae, given that this trait has been documented in other representatives of Bignoniaceae, in three clades defined by Olmstead *et al.* (2009): the Palaeotropical clade (Ogundipe and Wujek, 2004), Tecomeae (Parija and Samal, 1936) and Jacarandeae (Zatta *et al.*, 2009). This trichome morphotype is widely distributed in Bignonieae, but has only been reported in a few representatives of the family in general, probably reflecting the lack of detailed studies of trichome structure in other lineages of Bignoniaceae.

The density and size of the glandular heads represent the most variable features of *Pg* trichomes, differently from the other trichome types studied here. Ancestral state reconstructions of the individual features of this trichome type indicate that the MRCA of all Bignonieae probably had high densities of *Pg* trichomes with intermediate head size on the abaxial portion of leaflet blades. Even though the MRCA of the 'multiples of four' clade and '*Arrabidaea* and allies' clade (*sensu* Lohmann, 2006) had *Pg* trichomes with intermediate size heads and densities, the *Pg* trichomes increased in size and density in the first clade and decreased in size and density in the second one. Trichomes with large heads and high densities were relatively rare in Bignonieae, with huge large heads being restricted to two lineages (*Pyrostegia* and *Stizophyllum*), and huge high densities only present in 11 lineages, and being particularly common in *Amphilophium*.

*Stipitate glandular (*Sg*) trichomes.* This is the first record of *Sg* trichomes on the vegetative portions of representatives of Bignonieae. This kind of glandular trichome generally lacks a sticky secretion, and has been associated with alternative biotic defences or direct defence against herbivores. This trichome type has never been associated with ant attraction, which has more generally been associated with nectar-secreting trichomes. For example, adhesive glandular trichomes, a kind of *Sg* trichome, have been associated with spider bodyguards that often feed on carcases of insects that adhere to these trichomes (Morais-Filho and Romero, 2010). Ancestral character state reconstructions of this trichome morphotype indicate that *Sg* trichomes were not present in the MRCA of the group, independent of the plant portions analysed, and suggest that this trichome type has evolved at least nine times in the group. These results indicate that even though there is general structural similarity between the *Sg* trichomes encountered in the various lineages of Bignonieae, these similarities are not based on a common ancestry, suggesting a homoplastic pattern of evolution of these secretory structures.

*Patelliform/cupular glandular (*P/Cg*) trichomes.* The *P/Cg* glandular trichomes were observed on different plant parts, but only on the abaxial side of the leaflet blade in nearly all species of the group. These trichomes are associated with nectar secretion in Bignoniaceae, and have been commonly called extrafloral nectaries (EFNs) as they are not thought to be involved in pollination (Elias, 1983). However, the specific function of *P/Cg* trichomes has only been studied in five representatives of Bignoniaceae: *Campsis radicans* (Elias and Gelband, 1975), *Catalpa speciosa* (Stephenson, 1982), *Catalpa bignonioides* (Ness, 2003), *Anemopaegma album* and *Anemopaegma scabriusculum* (Nogueira *et al.*, 2012*a*). This terminology (EFNs) was widely used in Bignoniaceae in spite of the lack of functional evidence for this structure, i.e.'EFN' has been used in the literature solely to refer to a particular trichome morphotype, without direct evidence for its function.

In Bignonieae, *P/Cg* trichomes were documented in all 105 species sampled in the tribe, and also in the 16 species of Bignonieae trichome whose morphoanatomy was analysed in detail. Notably, this trichome type has been the most widely reported in all previous anatomical studies conducted in Bignoniaceae (Fig. 1). A mixture of terminologies has been used to designate these structures, obscuring the phylogenetic signal of this trichome type among species. For example, Seibert (1948) called *P/Cg* trichomes 'glands', whereas Elias (1983) called them 'extrafloral nectaries', terminologies that are more directly associated with the function than the overall morphological structure. On the other hand, terminologies that are more directly associated with the overall morphological structure of trichomes have also been proposed. For example, Díaz-Castelazo *et al.* (2005) treated *P/Cg* trichomes as 'scale--like trichomes', and Laroche (1974) called *P/Cg* trichomes 'patelliform glands'. In addition, Rivera (2000) used the topographical terminology proposed by Fahn (1979) and characterized *P/Cg* trichomes as belonging to 'type 12' (i.e. trichomes located in the outer epidermis of the calyx and/or corolla).

Our results corroborate earlier observations that *P/Cg* trichomes are more often present in the leaflet blades of representatives of Bignonieae (Elias, 1983). However, *P/Cg* trichomes have also been observed on the petals, sepals and fruits (Parija and Samal, 1936; Seibert, 1948; Laroche, 1974; Elias and Prance, 1978; Stephenson, 1982; Subramanian and Inamdar, 1989). The variation found in the structure of *P/Cg* trichomes of Bignonieae has been used to distinguish different species (Elias and Prance, 1978; Elias, 1983). Our study corroborates the earlier suggestion that the large and cupular morphotype of *P/Cg* that is marginally surrounded by the expansion of epidermis is typical of *Adenocalymma* ('volcano'-shaped glands as proposed by Lohmann, 2006). Furthermore, this study also corroborates the observation that the patelliform and cupular trichomes have a uniform anatomical structure (Rivera, 2000). Indeed, these variants of *P/Cg* trichomes have been shown to be generally homogeneous morphologically, varying only in terms of their shape and showing a differential cell proliferation and expansion, which is larger in cupular trichomes. These results thus suggest that cupular and patelliform trichomes are only superficially different and best treated under the same type of trichome called here *P/Cg* trichomes.

Ancestral character state reconstructions indicated that the MRCA of Bignonieae already bore *P/Cg* trichomes on both surfaces of the leaflet blades (most probably just on the abaxial side of leaflet blades). However, similar to what was found for *Pg* trichomes, this trait has also been documented in some representatives of other tribes/clades of Bignoniaceae (Parija and Samal, 1936; Seibert, 1948), indicating that their origin probably precedes the origin of the tribe. It has been

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hypothesized that *P/Cg* trichomes may simply represent an increase in complexity of *Pg* trichomes (Parija and Samal, 1936; Elias and Newcombe, 1979). In this case, the structural changes resulting in the evolution of these two trichome types would have occurred before the diversification of Bignonieae, since both types of trichomes were already present in the MRCA of the tribe. However, this hypothesis remains to be tested in a broader systematic context, in combination with detailed ontogenetic and evolutionary studies of both trichome types. Ontogenetic studies conducted with other representatives of Lamiales demonstrated an ontogenetic resemblance between the patelliform glandular trichomes of the Acanthaceae (McDade and Turner, 1997) and those of Bignoniaceae (Inamdar, 1969; Subramanian and Inamdar, 1986*a*, *b*). In both cases, the trichomes are formed from a single initial protodermic cell, suggesting that the trichomes found in both families might have a common origin, which would imply a more ancient origin for this structure.

Ancestral character state reconstructions of the density of *P/Cg* trichome morphotype per leaflet suggest that the ancestor of tribe Bignonieae probably had low densities of *P/Cg* trichomes on the adaxial side of leaflets and intermediate densities of trichomes on the abaxial side, with intermediate secretory head size. Furthermore, ancestral character state reconstructions also suggest that the ancestor of Bignonieae did not have *P/Cg* trichomes in the interpetiolar portions of stems nor in the prophylls of the axillary buds, with this feature evolving much more recently in other representatives of the tribe. Since these trichomes have been shown to be associated with ant attraction, and species with higher numbers of nectar-secreting trichomes had higher abundances of ants on the plants (Nogueira *et al.* 2012*a*, *b*), portions of the plant with high concentrations of these trichomes probably have a better protective role against plant herbivores. Thus, the clustering of such secretory structures on particular plant regions, as observed on the prophylls of axillary buds and interpetiolar portions of stems, suggests that these trichomes could be indirect protecting plant tissues that are costly to produce (i.e. the meristems of the axillary buds; Heil and McKey, 2003).

Concluding remarks

Despite the presumed functional importance of trichomes in different plant species and the importance for microevolutionary studies (Levin, 1973), little is still known about the macroevolutionary patterns of those structures in different plant families and on different plant organs. Traditionally, trichomes have been thought to evolve multiple times in the history of angiosperms. Theobald *et al.* (1979, p. 53) noted: 'it is evident that the trichomes have had infinite independent origins and as a result no 'phylogenetic' sequence, except on a very limited scale within certain small groups'. However, this does not seem to be the case in Bignonieae, at least in some parts of the plants, in which three major trichome morphotypes (non-glandular, peltate glandular and patelliform/cupular glandular) probably had a single origin in the group. Only stipitate glandular trichomes seem to have evolved multiple times independently of the position of their occurrence on vegetative organs. Our results indicate that most trichomes found in representatives of Bignonieae had the same origin and were similar morphologically, and are thus best treated under the same name (in each morphotype). A standardization of trichome terminology greatly facilitates comparisons among taxa, allowing inferences on relationships as well as inferences on their eco-evolutionary consequences.

Although we have improved the understanding of trichome evolution in a clade of angiosperms, it still remains unclear how trichomes interact ontogenetically and ecologically. For example, trade-off between trichome types could explain patterns of evolution in some clades (e.g. trade-off between *Sg* and *P/Cg* trichomes; Nogueira *et al.*, 2012*b*) or at least clarify the independent evolutionary trajectories of each trichome type. No information is available about the genetic basis of these trichome types and relationships with the patterns of position, density and size of trichomes. It is possible that putative heterotopic events (as discussed by Baum and Donoghue, 2002), with modifications of expression of some types of trichomes in specific plant parts, may have occurred during the evolutionary history of Bignonieae. Data directly related to the function of different trichomes are of importance for understanding the role of selection agents such as herbivores, sunlight and water for the evolution of trichome types. Experimental approaches that test functional hypotheses of trichomes in Bignonieae may be able to address the interpopulation variation of trichome characters in the future.

SUPPLEMENTARY DATA

Supplementary data are available online at www.aob.oxfordjournals.org and consist of the following. Table S1: taxonomic names of Bignoniaceae species used in anatomical/morphological studies presented in Fig. 1. Specimens: list of herbarium specimens of Bignonieae (Bignoniaceae) used in this study. Figure S1: ML ancestral character state reconstructions of non-glandular and *P/Cg* trichomes with presence/absence data of trichomes on different vegetative plant portions.

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